

Unconstrained Face Recognition: Establishing Baseline Human Performance via Crowdsourcing

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Unconstrained Face Recognition

- Identifying a person of interest based on unconstrained face imagery
- Challenges



2011 London riots



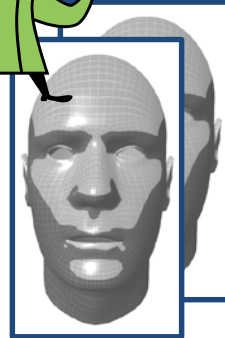
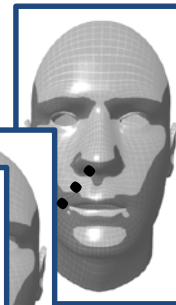
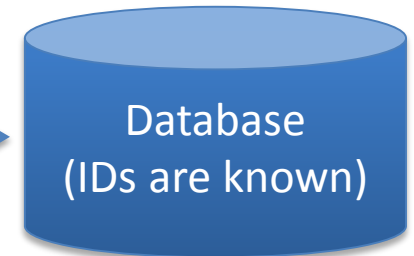
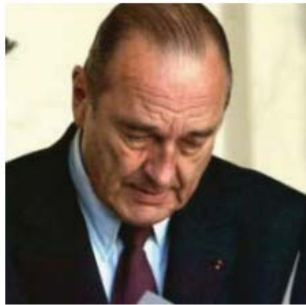
2013 Boston bombings



2014 Chicago robbery



“Human-in-the-Loop”



Top *K*
Matches



Important to analyze the accuracies achieved by both face matching algorithms and humans.

Unconstrained Face Databases

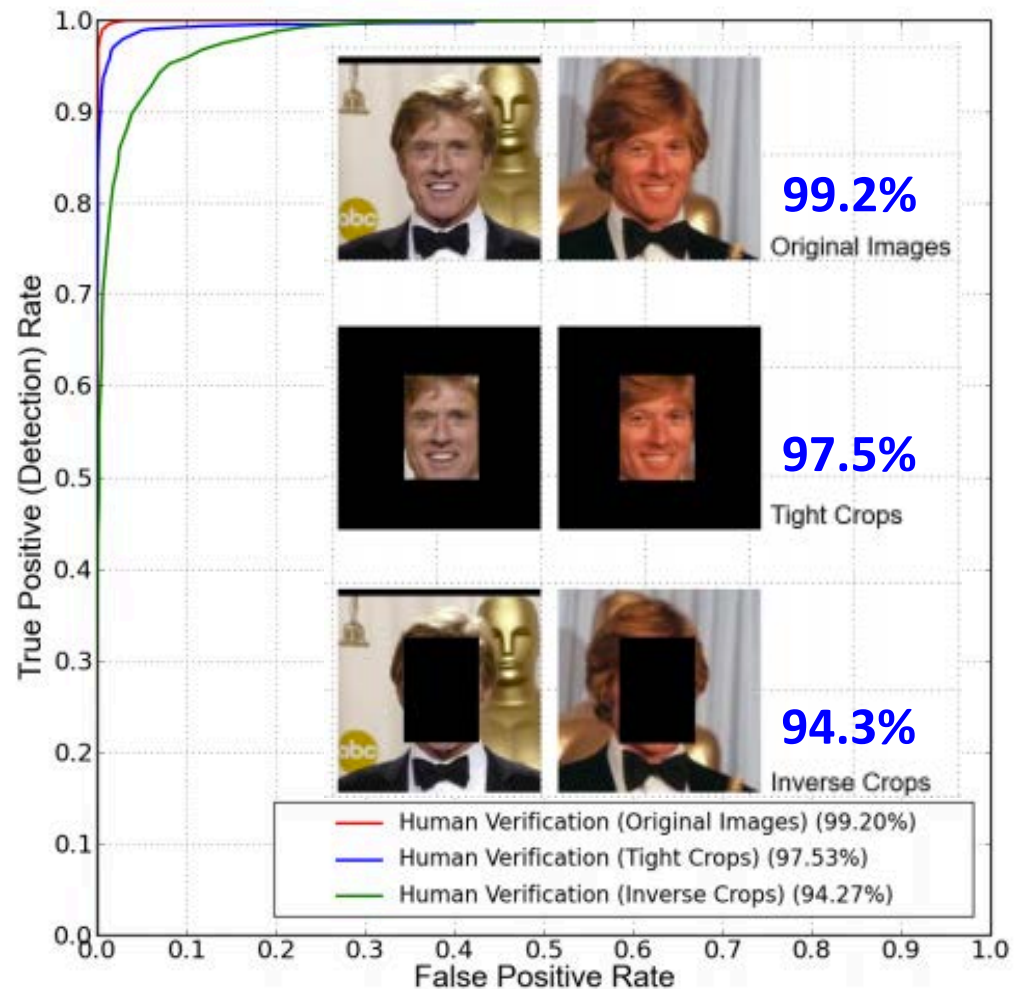
- **Labeled Faces in the Wild (LFW)**
 - 13,233 images of 5,749 people
- **YouTube Faces (YTF)**
 - 3,425 videos of 1,595 people
 - All subjects are also in LFW database
- **Experimental Protocols**
 - Face verification protocols
 - Work on LFW is extensive
 - Current performance: TAR > 99% at FAR = 1.0% (DeepFace)
 - Work on YTF is less extensive but gaining popularity



Prior Work on Human Performance

- Recent summary paper¹
 - FRVT 2006
 - FRGC
 - GBU
 - FOCS Video Challenge

- Kumar et al. on LFW² 



¹ P. J. Phillips and A. J. O'Toole. "Comparison of human and computer performance across face recognition experiments." *Image and Vision Computing*, 32(1):74-85, Jan. 2014.

² N. Kumar, A. C. Berg, P. N. Belhumeur, and S. K. Nayar. "Attribute and Simile Classifiers for Face Verification." *ICCV*, 2009.

Crowdsourcing on Amazon Mechanical Turk (MTurk)

- A large number of workers (a crowd) complete Human Intelligence Tasks (HITs) for requesters

Compare each pair of images. Please follow the directions below in the order that they are presented for each pair.

- First, is the same individual in both images? Pick the answer that best describes your decision.
- Second, is the face in either image familiar to you? If this statement is true, click the checkbox labeled "Familiar". If you know the individual's name, enter the name in the corresponding textbox. Can't remember the name? Enter any identifying information about the individual pictured, or leave the textbox blank.



Familiar? ☐

Name:



Familiar? ☐

Name:

Looking at the pair of images to the left, is the same person in both images?

- ☐ I am sure that they are the same.
- ☐ I think that they are the same.
- ☐ I cannot tell whether they are the same.
- ☐ I do not think that they are the same.
- ☐ I am sure that they are not the same.

Experimental Details

- Verification protocols
 - LFW: 6,000 face pairs of same vs. not-same
 - YTF: 5,000 face pairs of same vs. not-same
- Human responses are mapped to **confidence scores** 1 to 5 (similarity)
 - Human responses are **averaged** to obtain a smoothed score for each face image/video pair
 - 10 responses per pair for LFW
 - 20 responses per pair for YTF
 - Performance reported as ROC and accuracy of the binary decision (same vs. not same)

LFW Protocol Results

	TAR @ 1% FAR	TAR @ 10% FAR	Accuracy
Humans: Our Study	97.9	99.9	99.2
Humans: Kumar et al.	99.4	100.0	98.3
DeepFace: Taigman et al.	93.3	99.4	97.4
DeepID: Sun et al.	94.7	99.3	97.5
COTS	77.1	90.3	n/a

Data Collection Details

	LFW	
Total No. of Workers	307	
Avg. No. of HITs Completed Per Worker	39.1	

169 India
84 USA
20 other
34 blank

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169 India
84 USA
20 other
34 blank

Crowdsourcing on YTF Database

Compare each pair of videos. Please follow the directions below for each pair.

- **First:** View each video.
 - You can press the middle play button for each pair to start both videos simultaneously, or press each video to play them separately.
 - Is the same individual in both videos? Pick the answer that best describes your decision.
- **Second:** Is the face in either video familiar to you? If this statement is true, click the checkbox labeled "Familiar." If you know the individual's name, enter the name in the corresponding textbox. Can't remember the name? Enter any identifying information about the individual depicted, or leave the textbox blank.
- There are five pairs below.
- **IMPORTANT:** if any videos do not load correctly, please return this HIT. Thank you.



Familiar? ☐

Name:

[Play](#)



Familiar? ☐

Name:

Looking at the pair of videos, is the same person in both videos?

- ☐ I am sure they are the same.
- ☐ I think they are the same.
- ☐ I cannot tell whether they are the same.
- ☐ I do not think they are the same.
- ☐ I am sure they are not the same.

YTF Protocol Results

(Cropped Face Videos)

	TAR @ 1% FAR	TAR @ 10% FAR	Accuracy
Humans (USA)	80.6	96.7	89.7
Humans (India)	63.7	92.4	88.6
DeepFace: Taigman et al.	54.8	92.0	91.4
COTS	54.4	81.4	n/a

YTF Database Labeling Errors



(a)

(b)

(c)

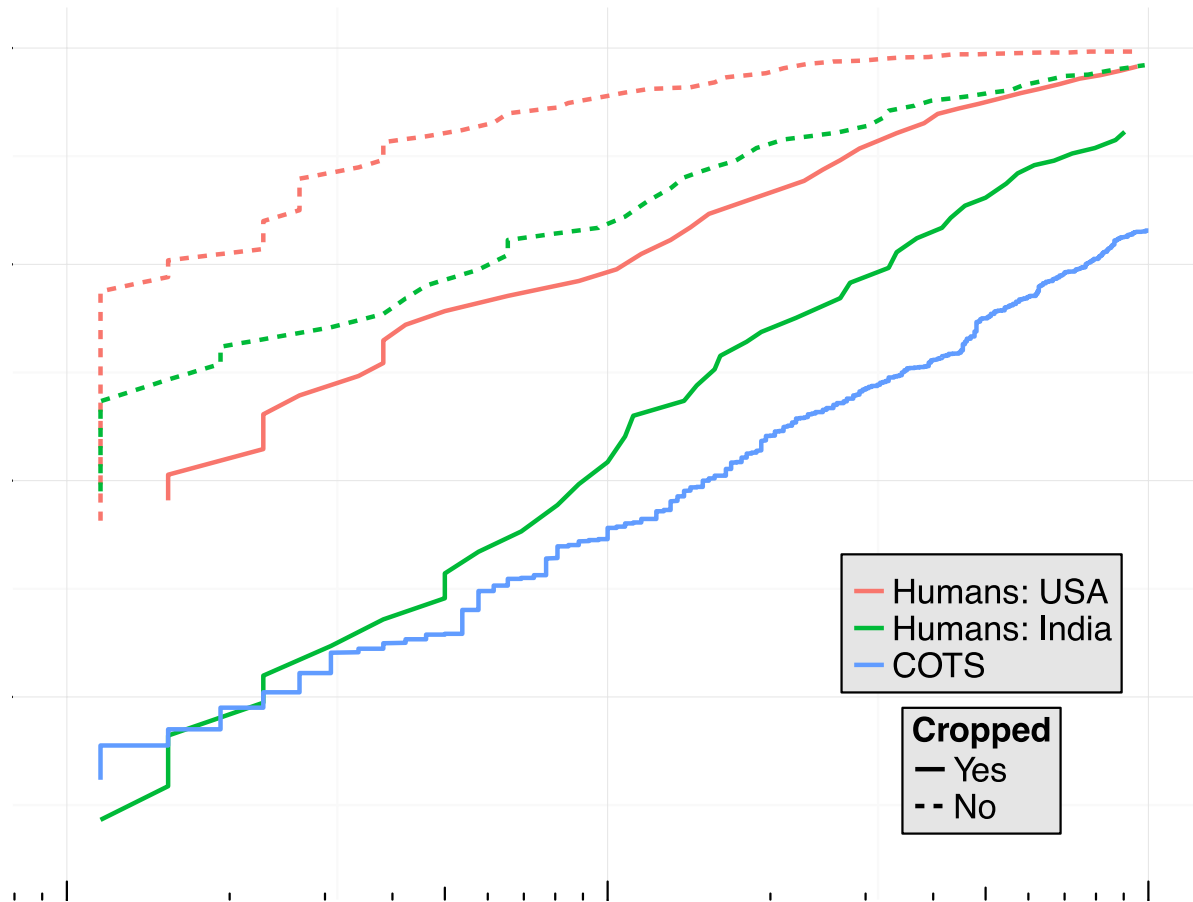
(d)

Pair	Original Label	Score	Correct Label
(a, d)	Genuine	1.55	Impostor
(a, c)	Genuine	3.75	—
(a, b)	Genuine	1.90	Impostor
(b, d)	Genuine	4.85	—
(b, c)	Genuine	1.50	Impostor
(c, d)	Genuine	1.35	Impostor

111 of 2,500 genuine face pairs in the YTF protocol are actually impostors.

YTF Results

(Original vs. Cropped Face Videos)



Context Assists with Recognition

@ FAR = 1%

(a) Genuine



Decision

Accept

Athletic uniform
helps in original
videos

(b) Impostor



Decision

Reject

Hair vs. no hair
helps in original
videos

Other-Race Effect?

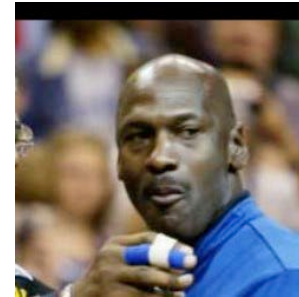
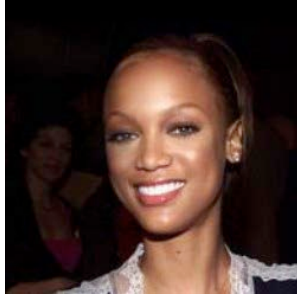


**62% of all subjects in the LFW database are White males.
4,350 White-to-White pairs, 168 Asian-to-Asian pairs in YTF protocol.**

Average accuracies (%) of individual MTurk worker responses for unfamiliar YTF face videos with respect to race demographics

Country	Study	White	Asian	non-w/A
USA	Original	86.0 ± 0.5	81.6 ± 2.2	81.9 ± 1.9
India	Original	83.0 ± 0.5	83.6 ± 2.4	79.5 ± 2.4
USA	Cropped	82.0 ± 0.5	77.5 ± 2.5	73.0 ± 2.3
India	Cropped	78.7 ± 0.6	77.7 ± 2.9	74.4 ± 2.5

Familiarity



Frequencies (%) of responses that were reported as familiar

	USA	India
Original	16.4	8.8
Cropped	11.1	1.2

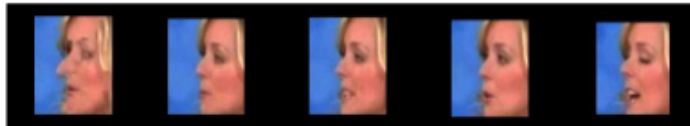
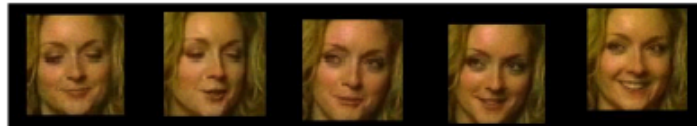
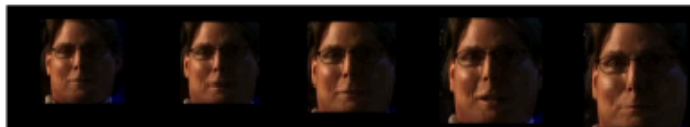
Average accuracies (%) of individual MTurk worker responses for YTF face videos

Country	Study	unfamiliar	familiar
USA	Original	86.3 ± 0.5	93.5 ± 0.3
India	Original	83.1 ± 0.6	87.8 ± 0.3
USA	Cropped	85.7 ± 1.3	92.6 ± 0.6
India	Cropped	80.4 ± 1.4	82.9 ± 0.4

Crowd Performance > COTS

Scores and Decisions
@ FAR = 1%

(a) Genuine Pairs

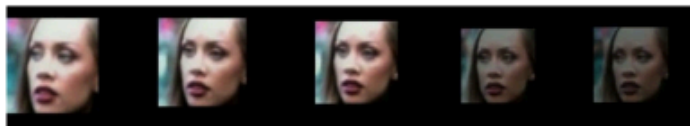
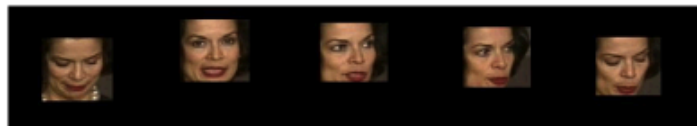


Human	COTS
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4.55 (14) <i>Accept</i>	0.12 <i>Reject</i>
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4.30 (6) <i>Accept</i>	0.09 <i>Reject</i>
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(b) Impostor Pairs



Human	COTS
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1.00 (0) <i>Reject</i>	0.36 <i>Accept</i>
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1.15 (4) <i>Reject</i>	0.44 <i>Accept</i>
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Single Human vs. Crowd Performance

- Randomly select a single response per face pair (100 times)

Country	Study	Rand. Ind. Acc.	Crowd Acc.
USA	Original	87.6 ± 0.4	95.8
India	Original	84.0 ± 0.5	93.7
USA	Cropped	83.2 ± 0.4	91.4
India	Cropped	79.0 ± 0.5	90.0

- Accuracies of the 20 humans who completed the most HITs for each YTF study

Country	Study	Num. HITs	Accuracy
USA	Original	$1,658 \pm 434$	89.1 ± 9.0
India	Original	$1,565 \pm 609$	84.7 ± 6.3
USA	Cropped	$1,900 \pm 367$	83.4 ± 4.9
India	Cropped	$3,621 \pm 1,005$	78.9 ± 10.1

Conclusions

- Human performance on face recognition can depend on country of origin of workers
 - Familiarity and/or other-race effect
- Machines are reaching “human” performance but...
 - Crowd response appears better than single human
 - The performance of trained face examiners is likely much higher than that measured via crowdsourcing
 - Examiners typically review the top K highest matches
- Crowdsourcing can be used to help verify the ground truth labels of a large database

Thank you!